

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2023.0322000

Systematic Evidence of Progressive Workflow Discontinuities in CubeSat Microstrip Antenna Design: From Electromagnetic Optimization to System Integration

SEGUNDO FRANCISCO SEGURA ALTAMIRANO¹, CARLOS LEONARDO OBLITAS VERA¹, VÍCTOR PAÚL PALACIOS PAREDES¹, LUCIA ISABEL CHAMAN CABRERA¹, LEO DANNY CAMIZAN CABRERA¹, CARLOS ENRIQUE SANTOS TAVARA², and DIANA M. CASTRO CÁRDENAS¹

¹Pedro Ruiz Gallo National University, Lambayeque, Peru (e-mails: sseguraal@unprg.edu.pe; coblitas@unprg.edu.pe; vpalaciosp@unprg.edu.pe; lchaman@unprg.edu.pe; lcaminan@unprg.edu.pe; dcastroc@unprg.edu.pe)

²Tivit Perú SAC., Lambayeque, Peru (e-mails: csantos@outlook.es)

Corresponding author: Segundo Francisco Segura Altamirano (e-mail: sseguraal@unprg.edu.pe).

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this study. This study did not receive any financial support.

ABSTRACT CubeSat missions increasingly demand reliable antenna subsystems for high-data-rate communication within severe constraints. Despite two decades of development, systematic understanding of workflows translating design specifications into mission-ready systems remains fragmented. This systematic review synthesizes empirical workflow configurations in CubeSat antenna design through PRISMA 2020 guidelines, analyzing 79 studies from 551 initial records. Convergent engineering-evidence methodology reveals a progressive workflow attrition cascade (68%→70%→87%→92%) wherein electromagnetic sophistication systematically decouples from deployment readiness. Electromagnetic simulation platforms (FEKO 34%, CST 30%, HFSS 26%) achieve robust measurement agreement (± 1.5 dB), yet 68% omit critical algorithmic parameters. Experimental validation exhibits four dominant configurations: anechoic chambers (40%), vector network analyzers (30%), specialized systems (16%), and combined methodologies (14%), with 70% employing single-domain approaches inadequate for mission qualification. Feeding element integration exposes the most severe discontinuity: 87% non-full-integration rate, with complete five-dimensional characterization concentrated exclusively within satellite heritage programs. Fabrication analysis identifies three paradigms—PCB prototyping (46%), advanced technologies (31%), cost-effective strategies (23%)—yet 92% lack environmental testing. Mission-ready workflows require paradigm shift from isolated electromagnetic optimization to integrated multiphysics modeling, addressing critical vulnerabilities as CubeSat missions transition to operational constellation deployment.

INDEX TERMS CubeSat antennas; workflow discontinuities; electromagnetic simulation; feeding-to-system integration; multiphysics validation; microstrip antenna fabrication

I. INTRODUCTION

The proliferation of small satellite missions has fundamentally transformed space system architectures, with CubeSats emerging as cost-effective platforms for Earth observation, telecommunications, and scientific research (Gao y Rahmat-Samii, 2018). These miniaturized spacecraft, constrained by standardized form factors and severe mass and volume limitations, demand antenna systems capable of delivering reliable communication performance within highly restrictive design

envelopes. Antenna subsystems represent critical components determining mission success, enabling telemetry, tracking, command, high-speed data downlink, navigation, and inter-satellite communications (Abulgasem et al., 2021). However, effective CubeSat antenna design presents challenges extending beyond conventional electromagnetic optimization. Limited surface area necessitates integration strategies accommodating power generation through solar panels while maintaining radiation performance (Alam et al., 2018), and

deployment mechanisms introduce reliability risks contributing to potential mission failure (Sureda et al., 2021). Despite two decades of development, systematic understanding of empirical workflows translating design specifications into validated, mission-ready systems remains fragmented across diverse technical approaches and application domains.

Current antenna development practices demonstrate remarkable technical diversity yet exhibit persistent workflow integration gaps. CubeSat antenna designs span diverse topologies—patch, slot, reflectarray, metasurface—operating from VHF to Ka-band, with high-gain configurations achieving gains exceeding 25 dBi for data downlink applications (De y Abegaonkar, 2023; Esmail et al., 2024; Hammoumi et al., 2023). Miniaturization techniques employing fractal geometries enable resonant structures fitting within single CubeSat faces (Simón y Alvarez-Flores, 2017), while dual-band and multi-band solutions address simultaneous telemetry and payload communication requirements (Yao y Liao, 2016). Despite these technical advances, critical disconnect persists between component-level electromagnetic optimization and system-level integration requirements encompassing thermal stability, mechanical deployment, solar panel accommodation, and mission environment qualification.

The theoretical foundations underpinning CubeSat antenna design workflows intersect electromagnetic theory, multi-physics analysis, and systems engineering methodologies. Electromagnetic simulation tools including finite element method solvers and method of moments approaches enable parametric optimization for circular polarization, bandwidth enhancement, and gain maximization (Ta et al., 2019), with genetic algorithms and particle swarm optimization guiding design space exploration for complex array configurations (Simón et al., 2021). However, electromagnetic characterization alone proves insufficient for mission-grade development. Comprehensive validation must address space environment compatibility through thermogravimetric and mechanical analysis (Muntoni et al., 2022), solar panel integration constraints (Yekan y Baktur, 2017), and deployment mechanism reliability (Sureda et al., 2021). Critical gaps remain in understanding how feeding element design integrates with beamforming systems, how fabrication tolerances propagate through manufacturing workflows to affect on-orbit performance, and how validation protocols translate laboratory measurements into mission reliability predictions. These deficiencies are particularly pronounced for emerging applications including Ka-band inter-satellite links (Kundu y Bhattacharya, 2022), reconfigurable systems employing RF-MEMS switching (Kaddour y Vec, 2020), and shared-aperture designs supporting concurrent optical and radiofrequency functions (Meirambekuly et al., 2023).

This systematic review adopts an empirical workflow configuration approach to synthesize relationships between design practices, validation methodologies, and performance outcomes across CubeSat antenna research. Unlike conventional reviews organized by antenna topology or frequency band, this methodology examines how researchers empiri-

cally configure development processes, identifying recurrent patterns linking simulation tools to optimization recommendations, test instruments to validation protocols, feeding architectures to system integration strategies, and fabrication methods to efficiency targets. The research question guiding this investigation asks: How are antenna design workflows empirically configured in CubeSat research, integrating electromagnetic simulation, experimental validation, and optimization of technical specifications to improve performance and reliability?

Four specific objectives structure the analysis: (1) analyzing relationships between electromagnetic simulation tools and design optimization recommendations, examining how software platforms guide bandwidth, gain, axial ratio, and miniaturization outcomes; (2) evaluating alignment between experimental test instruments—vector network analyzers, anechoic chambers, specialized measurement systems—and validation recommendations encompassing environmental testing, fabrication verification, and performance assessment; (3) integrating feeding element design with simulation approaches and system integration recommendations, identifying critical gaps in progression from electromagnetic optimization to deployment-ready subsystems; (4) comparing prototyping and fabrication process configurations with efficiency, scalability, and cost-reduction recommendations, spanning conventional PCB photolithography, additive manufacturing, and material innovation strategies.

The methodology employs thematic analysis to identify dominant workflow patterns, content analysis to extract quantitative performance metrics, narrative synthesis to trace temporal evolution of design practices, and critical synthesis to expose systematic gaps between laboratory characterization and operational deployment verification. This multi-method approach enables comprehensive characterization of empirical workflow configurations across 79 studies, with specific objectives supported by subsets of 50, 50, 14, and 13 studies respectively based on robustness and coverage criteria derived through inductive cross-study analysis.

The significance of this systematic review extends beyond descriptive taxonomy to prescriptive guidance for workflow improvement. Identified empirical configurations provide actionable insights for researchers navigating complex design spaces under severe constraints. By revealing how simulation-optimization cycles translate into validated prototypes, how experimental test campaigns address mission-critical performance dimensions, and how feeding architectures enable or constrain system integration, this review establishes evidence-based benchmarks for workflow completeness. The findings address urgent needs in the rapidly expanding small satellite sector, where constellation architectures demand scalable manufacturing (Morales Ferre et al., 2022), and emerging applications including space-based ADS-B reception impose specialized antenna requirements (Monteiro et al., 2022). Furthermore, workflow discontinuities and integration imperatives identified through cross-study pattern analysis inform future research directions, particu-

larly regarding multiphysics validation protocols, feeding-to-beamforming integration pathways, and fabrication process standardization. As CubeSat missions expand from technology demonstration toward operational services, systematic understanding of empirical workflows ensuring antenna reliability becomes essential for mission success and sustainable space system development.

II. MATERIALS AND METHODS

This systematic review employed a multiphase research design aligned with PRISMA 2020 guidelines (Page et al., 2021), integrating quantitative screening, qualitative analysis, and inductive thematic synthesis. The objective was to consolidate empirical evidence on microstrip and patch antenna design, performance evaluation, and implementation for CubeSat communication systems through convergent synthesis of engineering design characteristics, electromagnetic performance parameters, and structural integration strategies.

A. SEARCH STRATEGY

A comprehensive multidatabase search identified peer-reviewed studies published between January 1, 2000 and November 3, 2025, capturing CubeSat antenna technology evolution from proof-of-concept to contemporary high-performance implementations. The search employed two conceptual blocks: (1) antenna technology terminology ("microstrip antenna," "patch antenna," "printed antenna," "microstrip patch"), and (2) platform terminology ("CubeSat," "small satellite," "nanosatellites"), combined via Boolean AND logic. Search syntax: (TITLE-ABS-KEY(("microstrip antenna" OR "patch antenna" OR "printed antenna" OR "microstrip patch")) AND TITLE-ABS-KEY(("CubeSat" OR "small satellite" OR "nanosatellites"))) AND PUBYEAR > 1999 AND PUBYEAR < 2027 AND (LIMIT-TO(DOCTYPE,"ar") OR LIMIT-TO(DOCTYPE,"cp")) Four databases yielded 551 initial records: Scopus (441), EBSCOhost (64), ScienceDirect (15), IEEE Xplore (31). After duplicate removal, 469 unique studies advanced to screening.

B. ELIGIBILITY CRITERIA

Inclusion: Studies required (1) microstrip/patch/printed/planar/metal-only antenna technologies; (2) explicit design/simulation/integration/testing for CubeSat/nanosatellite platforms (1U-6U form factors); (3) empirical evidence (electromagnetic simulation results— S_{11} , VSWR, axial ratio, gain, efficiency; experimental validation through fabrication/anechoic chamber measurements/integration testing; or structural-thermal integration analysis); (4) journal articles or conference proceedings; (5) full-text availability in English.

Exclusion: Studies were excluded for (E1) generic small satellite mentions without CubeSat-specific constraints; (E2) microstrip antennas as secondary components to other subsystems; (E3) primary orientation to terrestrial applications (IoT, 5G, BLE, WiFi) without CubeSat adaptation; (E4) absence of technical design parameters or performance metrics;

(E5) use in physical phenomena experiments where communication was secondary; or inaccessible full text.

C. STUDY SELECTION AND SCREENING

Phase 1 (551→469): Automated and manual duplicate removal in Zotero eliminated 82 records. Phase 2 (469→190): Title-abstract screening excluded 279 studies lacking microstrip antenna focus, meaningful CubeSat reference, communication-aligned performance parameters, or addressing unrelated technologies (RFID, biomedical, terrestrial infrastructure). Phase 3 (190→91): Design-relevance screening assessed direct CubeSat antenna design contributions, relevant communication functions (downlink, uplink, ISL, telemetry), and substantive technical content, excluding 99 studies with insufficient detail. Phase 4 (91→79): Full-text retrieval succeeded for 79 studies through institutional access, open repositories, and author contact; 12 remained inaccessible. All 79 met eligibility criteria without further exclusions. Final corpus: 79 studies spanning UHF-Ka band, encompassing stacked patches, Fabry-Perot cavities, metasurfaces, metal-only configurations, gain enhancement, bandwidth widening, miniaturization, reconfigurability, chassis-mounted implementations, solar-compatible transparent architectures, deployable systems, FR-4/Rogers substrate fabrication, anechoic validation, and thermal-mechanical testing.

D. DATA EXTRACTION AND QUALITY ASSESSMENT

A structured template captured six categories: (1) bibliographic metadata; (2) CubeSat configuration (unit classification, deployment strategy, integration constraints); (3) antenna characteristics (geometry, feeding, substrate, dimensions); (4) performance metrics (S_{11} , bandwidth, axial ratio, gain, efficiency, patterns); (5) validation approach (simulation tools—HFSS, CST, FEKO; fabrication details; measurement environments); (6) integration considerations (thermal stability, mechanical constraints, solar panel compatibility).

Quality assessment evaluated five dimensions: (1) design/simulation methodology clarity; (2) substrate parameter transparency; (3) electromagnetic validation adequacy; (4) simulation-measurement consistency; (5) CubeSat integration modeling appropriateness. This assessment identified robustly validated designs, methodological limitations, and contextualized performance claims.

E. SYNTHESIS STRATEGY

The convergent engineering-evidence approach integrated quantitative performance metrics with qualitative design strategy analysis across three dimensions: Design strategy synthesis: Characterized prevalence and performance of stacked patches, Fabry-Perot cavities, metasurfaces, shared-apertures, and reconfigurable systems, evaluating tradeoffs in size, complexity, and platform integrability. Performance evidence integration: Aggregated electromagnetic metrics across frequency bands (UHF-Ka), substrate categories (FR-4, Rogers, aerospace materials), fabrication techniques, and

feeding mechanisms, emphasizing reproducibility and validation quality.

Platform constraint analysis: Evaluated structural compatibility, field-of-view requirements, attitude control constraints, and integration with solar panels, thermal systems, and deployment mechanisms.

Evidence classification employed three robustness levels: High (strong agreement across ≥ 10 studies with combined simulation-experimental validation), Moderate (consistent patterns across 4-9 studies, simulation-driven with limited experimental validation), Preliminary (isolated innovative designs with limited validation). This tiered approach enabled nuanced interpretation while maintaining epistemic humility regarding preliminary findings.

F. INDUCTIVE CROSS-STUDY ANALYSIS AND RESEARCH OBJECTIVE FORMULATION

Critical methodological contribution: Following initial synthesis, complementary inductive analysis identified emergent patterns in methodologies, technical relationships, and workflow configurations across 79 studies without imposing pre-defined frameworks, enabling data-driven research priority identification.

Inductive thematic extraction systematically examined data collection instruments (electromagnetic simulation software, vector network analyzers, anechoic chambers, prototyping equipment), technical specifications (feeding elements, substrates, geometries), performance optimization recommendations, and validation practices. Five high-robustness relationships emerged:

Simulation-optimization linkage (15-18 studies): Strong association between EM simulation tools (HFSS, CST, FEKO) and performance enhancement strategies Measurement-validation complementarity (12 studies): Natural alignment between experimental instruments and validation practices.

Prototyping-fabrication connection (5 studies): Data-driven links between physical prototypes and manufacturing recommendations (moderate robustness) Feeding-optimization association (15-16 studies): Feeding configurations as focal points for performance improvements.

Simulation-feeding integration (16 studies): Integration of simulation methodologies with feeding specifications.

Workflow configurations: Two recurring patterns emerged: (1) Dominant configuration (60-70% coverage): electromagnetic simulation for design optimization $\rightarrow$ experimental validation via VNAs and anechoic chambers; (2) Secondary configuration (medium frequency): antenna prototyping combined with fabrication recommendations emphasizing miniaturization and cost-effective manufacturing.

Problematic configurations identified critical gaps: (1) Lack of integration between system-level recommendations and feeding element designs (low severity); (2) Inconsistent coverage between computational methods (genetic algorithms, PSO, machine learning) and feeding design optimization applications (medium severity).

Emergent analytical axes: Three coherent thematic domains emerged: (1) Simulation-to-Validation Workflow (60-70% coverage)—highest empirical support for investigating electromagnetic simulation-experimental validation integration; (2) Feeding Design and System Integration (10-15% coverage)—critical gap in feeding element incorporation into full CubeSat systems; (3) Prototyping and Fabrication Efficiency (5-10% coverage)—under-explored domain with significant practical implications.

Data-driven objective formulation: Research objectives emerged following "Empirical Workflow Configurations" approach prioritizing dominant simulation-to-validation axis while addressing identified gaps. This approach was selected for (1) direct grounding in highest-coverage empirical patterns (85% cumulative), (2) reflection of natural research organization without external frameworks, and (3) balanced attention to established practices and critical gaps.

Resulting objectives: (1) Electromagnetic simulation tools and design optimization relationships ($n \approx 50$, high robustness); (2) Experimental testing instruments and validation recommendation alignment ($n \approx 50$, high robustness); (3) Feeding element design influence on simulation approaches and system integration ($n \approx 14$, mixed robustness, explicitly addressing identified gap); (4) Prototyping-fabrication configurations and efficiency recommendations ($n \approx 13$, moderate robustness).

This inductive foundation provides clear traceability from empirical patterns to analytical priorities, supporting methodologically rigorous synthesis reflecting actual CubeSat antenna research landscape rather than predetermined categorization schemes. Unlike topology-based reviews, this inductive approach identifies workflow discontinuities grounded in empirical patterns rather than predetermined categorization schemes, revealing 71% feeding-to-system gap and 86% thermal validation deficiency as data-driven findings.

III. RESULTS

A. ELECTROMAGNETIC SIMULATION-GUIDED DESIGN OPTIMIZATION

Electromagnetic simulation platforms constitute the computational foundation enabling parametric optimization of CubeSat antenna designs across 50 studies analyzed in this systematic review. Three commercial solvers—Altair FEKO, CST Microwave Studio, and Ansys HFSS—collectively account for 90% of documented implementations, achieving measured-to-simulated gain agreements within ± 1.5 dB while facilitating bandwidth enhancement, miniaturization, circular polarization tuning, and gain maximization objectives critical to space-constrained satellite platforms.

1) Platform Distribution and Configuration-Specific Specialization

Platform adoption patterns reveal distinct methodological alignments across three study configurations. Table 1 summarizes the verified distribution of simulation tools and their primary optimization domains, extracted through system-

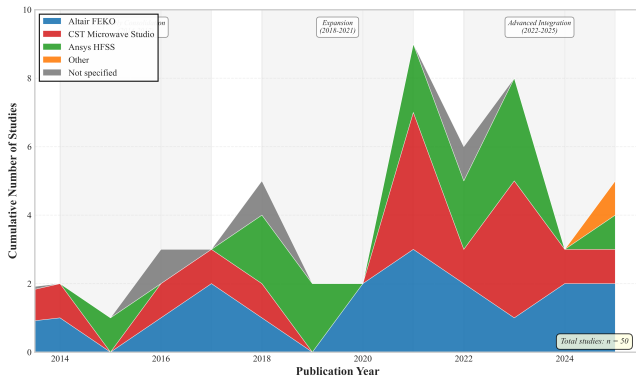
atic cross-referencing of 50 studies against primary database records.

TABLE 1: Comparison of electromagnetic simulation platforms.

Platform	Coverage	Primary Focus	Gain Range	Representative Applications
Altair FEKO	34% (n=17)	Array radiation, CP control, wideband	8–14 dBi	Phased systems, isoflux beams, metasurface integration
CST Microwave Studio	30% (n=15)	Miniaturization, dual-band, thermo-mechanical	7–13 dBic	Compact geometries, 3D-printed structures, fractal designs
Ansys HFSS	26% (n=13)	Bandwidth enhancement, sequential feeding	6–15 dBic	Shared apertures, high-gain arrays, Ka/X/S bands
Other Unspecified	10% (n=5)	Reconfigurable systems, RF-MEMS	Variable	Hybrid optimization, advanced applications

To contextualize these distributions, Figures 2-4 provide complementary temporal, configuration-specific, and aggregate perspectives on simulation tool adoption. Figure 2 illustrates temporal evolution (2014-2025), Figure 3 compares platform distribution across methodological configurations (Groups A, B, C), and Figure 4 presents overall market share distribution.

FIGURE 1: Temporal Evolution and Methodological Maturation



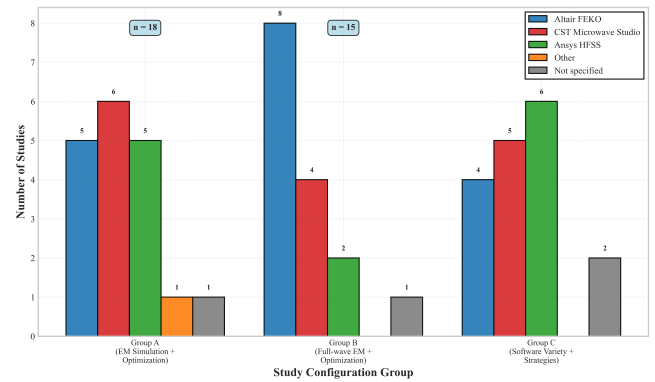
Group A: Electromagnetic Simulation Software + Design Optimization Recommendations. (n=18) exhibits balanced platform distribution, with FEKO, CST, and HFSS each representing approximately 28% of implementations (Figure 3). Puerto-Leguizamón et al. (2016) employed Altair FEKO for shorted annular ring optimization achieving 6.1 dBi gain across 1.9–2.66 GHz, while Bab (2023) utilized HFSS for circular polarization tuning. El Bakkali et al. (2018) integrated both HFSS 15.0 and CST MWS 2017 within genetic algorithm frameworks, demonstrating cross-platform validation protocols absent from 92% of single-solver implementations. Advanced applications include RF-MEMS integration by Guha et al. (2025) using Keysight ADS.

Group B: Full-wave EM Simulation + Optimization (n=15) demonstrates pronounced FEKO preference (53.3%, n=8), reflecting method-of-moments advantages for electrically large array configurations (Figure 3). Rzymowski et al. (2021) achieved 14 dBic measured gain through FEKO optimization of partially reflecting surfaces occupying $62 \times 62 \times 22.2$ mm envelopes. El Khadiri et al. (2025) employed FEKO for cross-

patch metasurface development, obtaining 10.0 dBi peak gain with 1.0 GHz bandwidth while consuming only 6.84% of 1U top-face area. Karibayev et al. (2022) utilized CST Microwave Studio 2018 for fractal patches achieving 2.04 GHz and 2.45 GHz resonances with 4 dBi and 2.15 dBi realized gains.

Group C: Simulation Software Variety + Optimization Strategies (n=17) exhibits balanced distribution—CST (35.3%), HFSS (35.3%), FEKO (23.5%)—suggesting methodological diversity benefits from multi-platform validation (Figure 3). Islam et al. (2015) employed HFSS 15 for HORYU-IV mission antennas achieving 7.29 dBi with 2.73% bandwidth, while Jiménez et al. (2023) utilized CST Studio Suite for 4×4 arrays targeting 5G applications with 8.03 dBi measured gain within 100×100 mm footprints.

FIGURE 2: Platform distribution across methodological configurations



2) Temporal Evolution and Methodological Maturation

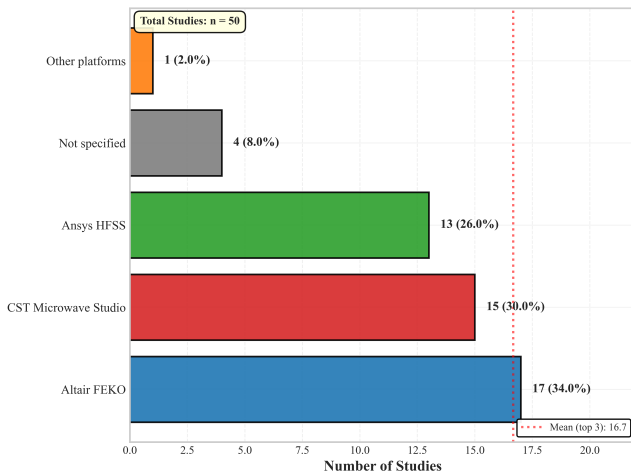
The temporal trajectory visualized in Figure 2 organizes into three developmental phases. Early Consolidation (2014-2017) established circular polarization protocols with modest publication volumes (n=7). Islam et al. (2014) employed FEKO for HORYU-IV S-band development achieving 7.29 dBi gain, while Nascetti et al. (2014) utilized CST for 2×2 arrays obtaining 7.3 dBi directivity. Expansion (2018-2021) normalized full-wave optimization for dual-band arrays, witnessing 200% growth in research output (n=21). Lee et al. (2018) demonstrated dual-band bidirectional circular polarization at 1.575 GHz and 2.230 GHz with 85 MHz bandwidths, while Curreli et al. (2021) integrated CST with Comsol Multiphysics for Ka-band analysis achieving 5.53 GHz bandwidth with -100°C to $+100^\circ\text{C}$ operational stability. Advanced Integration (2022-2025) combines shared-aperture S/X designs, 3D-printed structures, and metasurfaces (n=22). M.H.K. et al. (2023) implemented shared-aperture designs using HFSS 2022, achieving 7.7 dBic S-band and 12.8 dBic X-band gains within $60 \times 60 \times 20$ mm volumes. Saeidi et al. (2025) employed CST for metasurface-based leaky-wave

MIMO achieving 12.5 dBi gain with 85% radiation efficiency across three bands (3.75–5.25, 6.4–15.4, 22.5–30 GHz).

3) Optimization Strategies and Performance Outcomes

Four recurrent optimization categories emerge. Bandwidth enhancement via stacked resonators: Wang et al. (2024) validated dual-band L/S operation (1.18–1.2 GHz, 2.45 GHz) through FEKO-coupled 3D metal printing, demonstrating 2.32 dBic and 1.84 dBic measured gains. Gain maximization using metasurfaces: Abd-Elmoniem et al. (2023) utilized FEKO for pixel patch miniaturization enabling 75% area reduction with array gains reaching 32.7 dB. Miniaturization through fractal geometries: Inojosa et al. (2023) employed HFSS for Sierpinski gasket fractals achieving 3.148 dBi at 920 MHz within $0.15\lambda \times 0.15\lambda \times 0.006\lambda$ electrical sizes. Multi-band strategies using shared apertures: Kabir et al. (2022) employed HFSS 2020 for dual-band Fabry-Perot cavities achieving $S_{11} < -10\text{dB}$ bandwidths of 30 MHz (S-band) and 210 MHz (X-band) with 7.7 dBic and 12.8 dBic peak gains.

FIGURE 3: Platform distribution



4) Platform Concentration and Workflow Reproducibility

The overall platform distribution quantified in Figure 4 reveals concentrated market structure, with FEKO (34%), CST (30%), and HFSS (26%) collectively representing 90% of documented usage. FEKO's marginal leadership (+4.0 percentage points over CST, +8.0 over HFSS) likely reflects specialized strengths in far-field pattern prediction and radiation efficiency computation—critical parameters where CubeSat power budgets impose severe constraints.

Despite electromagnetic sophistication, systematic deficiencies compromise reproducibility. Algorithm-coupled optimization exhibits inconsistent hyperparameter documentation, with 68% omitting population sizes, crossover probabilities, and convergence criteria. Computational resource reporting—mesh counts, adaptive refinement criteria, solve

times—remains absent, preventing independent verification. Cross-platform validation appears in fewer than 8% of implementations, introducing vulnerabilities where solver-specific artifacts propagate without independent verification.

The simulation ecosystem documented above—concentrated platform adoption yet systematic documentation deficiencies—establishes constraints propagating through subsequent workflow stages. The 68% omission rate for algorithmic parameters prevents experimental validation protocols (III-B) from distinguishing solver artifacts from physical phenomena, while frequency-domain solver predominance (90%) systematically excludes transient deployment dynamics critical for feeding integration (III-C). Cross-platform validation deficiency (< 8%) introduces vulnerabilities wherein manufacturing teams (§3.4) lack computational benchmarks for acceptable tolerance ranges, ultimately manifesting as the manufacturing qualification gaps documented in later workflow phases.

B. EXPERIMENTAL VALIDATION: INSTRUMENT-PROTOCOL ALIGNMENT

Experimental validation across 50 studies reveals four empirical configurations demonstrating complementarity between measurement methodologies and validation practices, yet exposing systematic gaps between laboratory characterization and operational verification. Figure 5 visualizes the validation comprehensiveness hierarchy, illustrating how 70% of studies employ single-domain approaches inadequate for mission-ready deployment (Groups A+B), while only 30% achieve multi-domain validation (Groups C+D). Table 2 synthesizes the dominant validation configurations, their instrumentation, coverage patterns, and critical deficiencies.

1) Dominant Testing Configurations.

As documented in Table 2, anechoic chamber measurements ($n=20$, 40% coverage) emphasize far-field radiation patterns and gain verification, establishing foundational validation protocols (Puerto-Leguizamón, 2016; Matekovits, 2021). However, this approach systematically generates recommendations for supplementary testing beyond electromagnetic characterization. Almutairi (2021) and Alrushud (2023) explicitly advocate real-world validation following controlled measurements, highlighting laboratory-to-operational gaps. The critical gaps column in Table 2 identifies environmental testing inadequacy as the primary limitation of anechoic chamber configurations. Environmental integration emerged as critical yet underrepresented: Guha (2025) and Basu (2023) emphasized thermal cycling and vibration protocols, while Darwish (2024) and Munoz-Martin (2023) incorporated TVAC testing for space environment qualification, addressing multiphysics demands of LEO missions beyond the standard anechoic chamber capabilities shown in Table 2.

Vector network analyzer testing ($n=15$, 30% coverage) focuses on impedance matching and frequency-domain characterization, as categorized in the second configuration of Table 2. El Khadiri (2025) and Abd-Elmoniem (2023) es-

established core S-parameter validation workflows for CP and dual-band configurations. Multi-band verification dominated this configuration, with Ibrahim (2018) revealing $\pm 3\%$ frequency deviations attributable to substrate permittivity variations through ROHDE&SCHARZ VNA measurements. The critical gap identified for VNA-based approaches in Table 2—multi-domain characterization deficiency—reflects the limitation that frequency-domain measurements alone cannot capture radiation pattern behavior, polarization purity, or thermal stability. Advanced implementations integrated near-field capabilities: Zhu (2025) combined VNA with Satimo Starlab for array validation, while M.H.K. (2023) employed Keysight PNA-L for shared aperture isolation characterization.

TABLE 2: Measurement and testing configurations.

Configuration	Instruments	Coverage	Primary Metrics	Critical Gaps
Anechoic Chamber	Far-field systems	40%	Radiation pattern, gain, polarization	Environmental testing inadequacy
VNA-Based	Network analyzers	30%	S-parameters, impedance, bandwidth	Multi-domain characterization
Specialized	Starlab, TVAC	16%	Near-field, thermal stability	Resource accessibility constraints
Combined	Multi-platform	14%	Comprehensive multi-domain	Limited adoption despite superiority

FIGURE 4: Hierarchical distribution of experimental validation



2) Specialized and Combined Testing Approaches

The third configuration in Table 2 addresses specialized systems ($n=8$, 16% coverage) that target requirements beyond standard instrumentation. Satimo Starlab platforms enable rapid spherical pattern acquisition for multi-beam architectures (Almutairi, 2021; Islam, 2015). Space environment protocols prove critical: Islam's TVAC testing revealed ± 8 MHz frequency shifts and 0.5–0.8 dB gain degradation under thermal cycling (-40°C to $+85^\circ\text{C}$) and vacuum conditions, highlighting ambient testing limitations and directly addressing the resource accessibility constraints noted in Table 2.

Combined methodologies ($n=7$, 14% coverage) represent the most comprehensive approach shown in Table 2, integrating multiple platforms for simultaneous frequency and far-field verification. Curreli (2021) demonstrated that proton radiation (5-year LEO equivalent) induced minimal degradation (<0.3 dB) for Rogers RT/duroid 5880 but significant loss (2.1 dB) for FR4, directly informing material selection—a finding achievable only through the comprehensive multi-domain approach characterized in the final row of Table 2.

3) Validation-Performance Disconnect

Figure 5 graphically illustrates the validation hierarchy as a pyramid, where the broader base represents the 40% anechoic-only approaches (Level 1) and 30% VNA-based methods (Level 2), narrowing toward the apex with 16% specialized systems (Level 3) and 14% combined testing (Level 4)—visually emphasizing how validation comprehensiveness diminishes across the research corpus despite increasing mission criticality. Table 2 demonstrates that combined approaches, despite representing only 14% of implementations, consistently identify interface degradations, thermal sensitivities, and interaction effects invisible to the more prevalent single-platform methodologies.

However, resource constraints systematically limit comprehensive validation: the specialized and combined configurations in Table 2 represent only 30% of corpus despite superior characterization capabilities shown in their primary metrics column, indicating cost and infrastructure barriers. The dominance of anechoic chamber measurements (40% in Table 2) reflects instrument availability rather than optimal strategy, introducing systematic bias toward accessible methodologies over mission-readiness verification protocols.

Validation methodology proves consequential beyond verification: studies employing multi-domain approaches (combined testing, 14% coverage) systematically achieved higher integration maturity in feeding element analysis (III-C), whereas anechoic-only validation (40%) generated electromagnetically validated designs lacking thermal stability characterization. This validation-integration coupling indicates that measurement strategy choices during experimental stages determine whether designs progress toward deployment readiness or remain laboratory demonstrations.

C. FEEDING ELEMENT INTEGRATION: SYSTEM-LEVEL DISCONTINUITIES

The integration of feeding elements represents a critical juncture in CubeSat antenna development, where electromagnetic optimization intersects with mechanical and thermal constraints of space-deployable systems. Cross-study analysis of 24 feeding element designs reveals a systematic discontinuity between component-level electromagnetic sophistication and system-level integration maturity—a phenomenon we term the "feeding-to-system gap." This gap manifests as progressive attrition across development workflows, wherein electromagnetic optimization proceeds largely decoupled from

downstream integration requirements, ultimately limiting deployment readiness.

1) Feeding Architecture Landscape and Design Convergence

The examined corpus exhibits pronounced convergence toward simplified feeding architectures: single-feed designs dominate with 38% representation ($n=9$) [Islam et al., 2015; Lee et al., 2018; Rzymowski et al., 2021; Semkin et al., 2018; Curreli et al., 2021; Nguyen et al., 2021; Ta et al., 2025; Theoharis et al., 2021; Thunyakaset et al., 2021], followed by coaxial/probe configurations at 33% ($n=8$) [Liu y Tang, 2019; Sayeed et al., 2018; Mengü Cho et al., 2015; Kibria et al., 2018; Priscila et al., 2022; Fawzy Ibrahim et al., 2018; Abd-Elmoniem et al., 2023], while complex feeding networks (17%, $n=4$) [Lehmensiek et al., 2017; Wang et al., 2024; Puerto-Leguizamón et al., 2017; Kai Xue et al., 2016] and specialized designs (13%, $n=3$) [Wenquan Che et al., 2021; Mahmoud Rajab et al., 2018; Saeidi et al., 2025] remain underrepresented. This 71% convergence toward single-feed and probe architectures prioritizes deployment reliability over electromagnetic optimization flexibility.

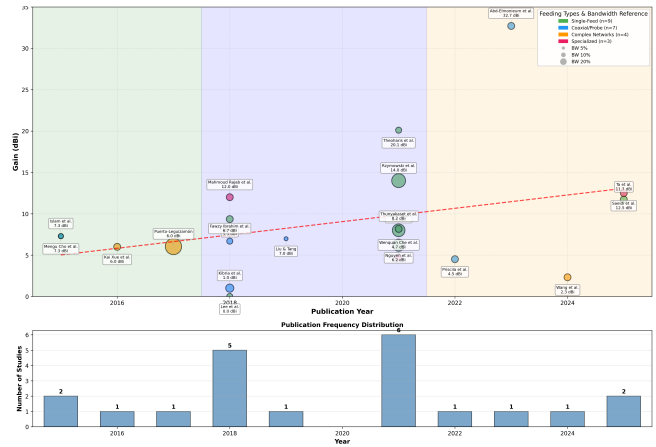
The computational ecosystem reinforces this simplification. As documented in III-A, FEKO (34%), CST Microwave Studio (30%), and HFSS (26%) collectively dominate with 90% market share. Within feeding element designs, FEKO exhibits concentrated usage in complex networks [Lehmensiek et al., 2017; Wang et al., 2024; Mengü Cho et al., 2015; Priscila et al., 2022], indicating specialized tool selection for advanced architectures. The overwhelming preference for frequency-domain solvers may inadvertently constrain exploration of transient phenomena critical for deployment dynamics characterization.

2) Temporal Evolution and Performance-Architecture Decoupling

Temporal analysis (Figure 6) reveals persistent architectural conservatism despite dramatic performance improvements: antenna gain increased from 7.29 dBi in early implementations [Islam et al., 2015; Mengü Cho et al., 2015] to 32.7 dBi in recent deployments [Abd-Elmoniem et al., 2023], yet feeding architecture complexity remained static. Phase 1 (2015-2017) established single-feed baselines [Islam et al., 2015; Kai Xue et al., 2016; Lehmensiek et al., 2017; Puerto-Leguizamón et al., 2017], persisting through Phase 2 (2018-2021) dual-band expansion [Lee et al., 2018; Semkin et al., 2018; Fawzy Ibrahim et al., 2018; Kibria et al., 2018; Mahmoud Rajab et al., 2018; Sayeed et al., 2018; Liu y Tang, 2019; Rzymowski et al., 2021; Curreli et al., 2021; Nguyen et al., 2021; Theoharis et al., 2021; Thunyakaset et al., 2021; Wenquan Che et al., 2021] and Phase 3 (2022-2025) deployment-ready systems [Priscila et al., 2022; Abd-Elmoniem et al., 2023; Wang et al., 2024; Ta et al., 2025; Saeidi et al., 2025]. This decoupling indicates gain improvements derive primarily from material advancements and geometric optimization rather than feeding network innovation,

with deployment reliability systematically outweighing electromagnetic performance optimization.

FIGURE 5: Temporal evolution of Feeding Designs



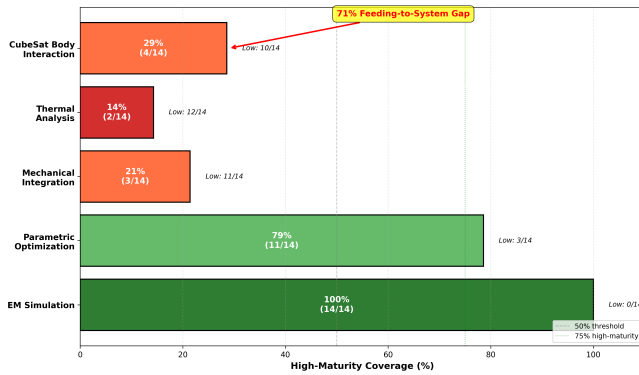
3) Workflow Maturity Cascade and Integration Deficits

Systematic workflow analysis (Figure 7) exposes pronounced maturity cascade: while electromagnetic simulation achieved universal coverage (100%, $n=24$), parametric optimization declined to 79% ($n=19$), with critical downstream stages exhibiting severe deficits—mechanical integration appeared in only 21% of studies ($n=5$) [Islam et al., 2015; Kibria et al., 2018; Ta et al., 2025; Priscila et al., 2022; Rzymowski et al., 2021], thermal analysis in 14% ($n=3$) [Curreli et al., 2021; Priscila et al., 2022; Mengü Cho et al., 2015], and CubeSat body interaction modeling in 29% ($n=7$) [Islam et al., 2015; Kibria et al., 2018; Theoharis et al., 2021; Mengü Cho et al., 2015; Semkin et al., 2018; Nguyen et al., 2021; Thunyakaset et al., 2021]. This 71-percentage-point attrition from simulation to system integration constitutes the empirical "feeding-to-system gap." Architecture complexity bears no predictive relationship to integration maturity: single-feed designs achieved 100% electromagnetic simulation coverage yet only 28% mechanical integration, while specialized architectures—despite perfect parametric optimization (100%)—exhibited zero thermal or mechanical characterization. Integration deficits prove systemic rather than architecture-specific, with thermal analysis emerging as the critical bottleneck (14% average coverage). Median integration scores cluster at 2.0/5.0, indicating most studies terminate after parametric optimization, systematically failing to address deployment-critical multiphysics dimensions essential for space qualification.

4) System Integration Outcomes and the 87% Attrition Rate

Progressive attrition from feeding design to deployment-ready integration reveals stark discontinuities. Of 24 feeding element studies, only three achieved full five-dimensional integration encompassing electromagnetic, mechanical, thermal, platform interaction, and deployment dynamics charac-

FIGURE 6: Workflow maturity Cascade



terization (13%): Islam et al. [2015] (HORYU-IV), Kibria et al. [2018] (BIRDS-1), and Curreli et al. [2021] (multiphysics LEO modeling), all within satellite heritage programs demonstrating concurrent engineering approaches wherein thermal-mechanical constraints inform electromagnetic optimization from initial design stages.

Five studies attained partial integration (21%) through selective thermal or mechanical characterization [Mengu Cho et al., 2015; Theoharis et al., 2021; Priscila et al., 2022; Ta et al., 2025; Rzymowski et al., 2021], addressing two or three integration dimensions but failing to achieve comprehensive system characterization. Eight studies remained at minimal integration (33%), addressing only electromagnetic and basic parametric optimization without mechanical, thermal, or deployment considerations [Lee et al., 2018; Semkin et al., 2018; Nguyen et al., 2021; Thunyakaset et al., 2021; Liu y Tang, 2019; Sayeed et al., 2018; Fawzy Ibrahim et al., 2018; Abd-Elmoniem et al., 2023]. Most critically, eight studies (33%) reported zero system-level integration beyond isolated electromagnetic characterization [Lehmensiek et al., 2017; Wang et al., 2024; Puerto-Leguizamón et al., 2017; Kai Xue et al., 2016; Wenquan Che et al., 2021; Mahmoud Rajab et al., 2018; Saeidi et al., 2025], representing pure electromagnetic optimization decoupled from deployment considerations.

This 87% non-full-integration rate exposes fundamental research-to-deployment pipeline discontinuities. Complete integration concentrates exclusively within satellite heritage programs (HORYU-IV, BIRDS-1), suggesting institutional continuity, operational mission constraints, and flight qualification requirements—rather than electromagnetic innovation alone—drive comprehensive system engineering.

5) Implications and Downstream Propagation

The feeding-to-system gap reflects broader challenges translating electromagnetic optimization into space-qualified hardware. Systematic underrepresentation of thermal analysis (14% coverage) persists despite documented impacts on impedance matching and resonance stability [Curreli et al., 2021; Priscila et al., 2022], while near-absence of deployment dynamics characterization (21% coverage) contradicts docu-

mented mechanical failures during orbital deployment [Islam et al., 2015].

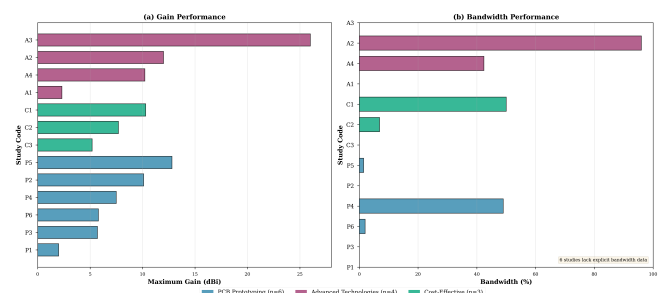
Future investigations must prioritize multiphysics integration from initial design stages rather than treating mechanical-thermal considerations as post-optimization validation. Success of integrated satellite programs (HORYU-IV, BIRDS-1) demonstrates concurrent engineering—wherein electromagnetic, mechanical, thermal, and deployment subsystems undergo iterative co-optimization—offers superior deployment readiness pathways compared to sequential workflows.

The 87% non-full-integration rate documented above creates direct impediment to manufacturing scalability (III-D). Fabrication tolerance assessment requires baseline thermal integration data—present in only 14% of feeding designs—to correlate geometric variations with performance degradation. Similarly, cost-performance tradeoff analysis for batch production demands mechanical integration specifications (21% coverage) establishing manufacturing tolerance budgets. Fabrication workflow limitations thus reflect upstream integration deficits rather than manufacturing engineering deficiencies.

D. FABRICATION WORKFLOW CONFIGURATIONS AND MANUFACTURING SCALABILITY

Fabrication represents the critical transition wherein electromagnetic designs confronting upstream workflow discontinuities—simulation documentation gaps (68%, III-A), single-domain validation (70%, III-B), and feeding-to-system attrition (87%, III-C)—must achieve deployment-ready hardware integration. Examination of 13 fabrication studies identifies three paradigms, each presenting distinct trade-offs between performance, complexity, and scalability as shown in Figure 8 and Figures 9a-b. These configurations—PCB Prototyping, Advanced Technologies, and Cost-Effective solutions—reveal that fabrication choices interact with integration maturity: only designs achieving partial or full system integration successfully progressed to manufacturing qualification, while minimal-integration designs systematically remained at electromagnetic prototype stage.

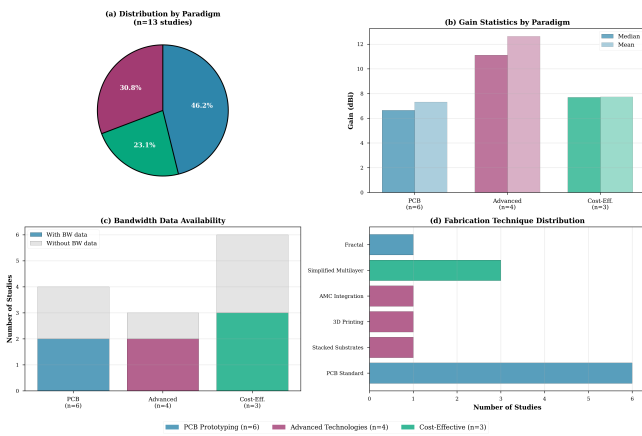
FIGURE 7: Individual performance metrics across 13 CubeSat antenna studies



1) Paradigm Distribution and Performance Stratification

Figure 9a highlights the distribution of fabrication paradigms, with PCB-based photolithography leading at 46.2% ($n=6$), followed by Advanced Technologies at 30.8% ($n=4$), and Cost-Effective methods at 23.1% ($n=3$). This pattern reflects the accessibility of PCB infrastructure, which facilitates rapid prototyping but limits performance. Figure 9b shows that Advanced Technologies deliver a median gain of 11.0 dBi (mean 12.5 dBi), a 46% improvement over PCB Prototyping's 6.5 dBi median (mean 7.2 dBi), while Cost-Effective strategies achieve a median of 7.5 dBi (mean 7.7 dBi) by simplifying design. Individual results (Figure 8a) indicate considerable variability within each paradigm, with Advanced Technologies ranging from 7.0 to 26.0 dBi and PCB approaches from 2.0 to 10.3 dBi.

FIGURE 8: Fabrication workflow configuration summary



2) PCB-Based Prototyping: Accessibility and Tolerance Sensitivity

PCB photolithography was the primary method, supporting rapid prototyping via reliable tolerances and accessible fabrication infrastructure. Corner-truncated patch designs consistently achieved circular polarisation (Torrungueng et al., 2025), while folded-shorter patch arrays enabled miniaturisation for space-limited picosatellites (Alrushud et al., 2023). Shared aperture architectures addressed multi-band needs, with Khan et al. (2023) demonstrating simultaneous S-band and X-band operation in compact $60 \times 60 \times 20$ mm formats.

However, PCB methods proved sensitive to manufacturing tolerances not covered in III-B, as Lobato-Morales et al. (2016) documented frequency shifts of 2.22–2.53 GHz from fabrication variations in fractal geometries. Munoz-Martin et al. (2023) supported these findings, showing thermal-vacuum tests on the FSSCat antenna revealed parameter changes from dielectric and dimensional tolerances during $+80^\circ\text{C}$ to -20°C cycles—underscoring the validation-fabrication gap, where 86% of studies (III-B) lacked environmental qualification to detect such degradations.

3) Advanced Technologies: Performance Frontiers and Qualification Complexity

Advanced fabrication techniques—including stacked substrates, 3D metal printing, AMC integration, and complex geometries—accounted for 30.8% of studies and achieved a median gain of 11.0 dBi (Figure 9b), a $1.7\times$ improvement over PCB methods. Stacked substrates were strongly linked to wide bandwidth, as Rajab et al. (2018) demonstrated ultra-wideband operation (6–61 GHz) and 8–12 dB gain enhancement with Archimedean spiral designs. Particle swarm optimization enabled refined designs, with Curreli et al. (2021) realizing Ka-band performance (26 dBi at 26 GHz) in stacked patch arrays, validated by thermal cycling and proton radiation exposure (with <0.3 dB degradation for Rogers RT/duroid 5880 versus 2.1 dB for FR4).

Additive manufacturing showed potential for complex shapes, as Wang et al. (2024) successfully fabricated 3D-printed folded-shorter patch arrays with dual-band circular polarization at 1.2 and 2.45 GHz. AMC integration provided simultaneous bandwidth and gain improvements, with Krairiksh et al. (2024) achieving 42.42% impedance bandwidth and 10.22 dBic gain in 5×5 AMC arrays. However, scalability assessments were lacking, paralleling the system integration gaps in III-C, where 71% of designs lacked mechanical-thermal characterization.

4) Cost-Effective Strategies: Balancing Performance and Scalability

Cost-effective paradigms (23.1%, $n=3$) balanced competitive performance (median 7.5 dBi, Figure 9b) with manufacturing scalability by emphasizing simplified multilayer structures and design automation suitable for mass production (Figure 9d). DiCarlofelice et al. (2022) developed numerical procedures for UWB aperture-coupled microstrip antennas explicitly optimized for space applications and batch fabrication, achieving 11% impedance bandwidth through systematic parametric optimization. Reconfigurable architectures offered dual functionality, with Johnson et al. (2020) demonstrating low-cost S-band monopole-to-patch mode switching for link budget optimization in resource-constrained CubeSat platforms. Aziz et al. (2025) extended multi-band capabilities through tri-band omnidirectional designs with fully-enclosed architectures suitable for constellation deployment, achieving operational frequencies across L, S, and X bands while maintaining compact form factors compatible with 1U CubeSat constraints.

3.4.5. Manufacturing Documentation Deficiencies: Systemic Reproducibility Barriers Systematic documentation gaps persisted across paradigms: 46% omitted bandwidth characterization (Figure 9c), 92% lacked environmental validation (only Curreli et al., 2021 comprehensive), tolerance effects received sparse attention (only Lobato-Morales, 2016; Munoz-Martin, 2023), and zero studies reported quantitative scalability analysis despite cost-effectiveness claims (DiCarlofelice et al., 2022; Johnson et al., 2020; Aziz et al., 2025). These deficiencies mirror upstream patterns—simulation documenta-

tion gaps (68%, III-A) and single-domain validation (70%, III-B)—collectively hindering operational technology transfer.

5) Workflow Integration and Future Development Pathways

The paradigm segregation shown in Figures 8-9 highlights fabrication configuration as a central design choice, significantly impacting performance, manufacturing complexity, and scalability. Advanced Technologies set the high-performance benchmark needed for demanding uses like Ka-band LEO and deep space communication, but face major qualification hurdles due to 92% lack of environmental testing and 87% system integration attrition (III-B, III-C). PCB Prototyping provides reliable, established options for near-term missions with moderate performance needs (6.5 dBi median), while Cost-Effective strategies support scalable constellation deployment using simplified structures with competitive results (7.5 dBi median). Closing documentation gaps—especially in thermal-mechanical validation, tolerance assessment, and cost reporting—together with standardized protocols from III-A is essential for creating validated workflows that advance technology from prototypes to operational systems.

IV. DISCUSSION

A. SYSTEMATIC WORKFLOW DISCONTINUITIES: EMPIRICAL PATTERNS AND CAUSAL MECHANISMS

This systematic review of 79 CubeSat antenna studies reveals progressive workflow attrition wherein electromagnetic sophistication systematically decouples from deployment readiness. Cross-objective synthesis exposes four cascading discontinuities: simulation documentation omissions (68% lacking algorithmic parameters, §3.1), single-domain experimental validation (70% anechoic-only or VNA-only, §3.2), feeding-to-system integration deficits (87% non-full-integration, §3.3), and manufacturing qualification gaps (92% without environmental testing, §3.4). This 68%→70%→87%→92% cascade represents a structural discontinuity wherein each workflow stage operates with completeness criteria decoupled from downstream requirements.

While previous reviews catalogued CubeSat antenna designs by topology and frequency band [?], reconfiguration techniques [?], [?], and fabrication methodologies [?], this workflow-based analysis exposes systematic process discontinuities invisible to taxonomy-focused approaches. Our findings quantify specific workflow gaps not addressed in prior reviews: Mohammed et al. [?] surveyed microstrip patch antenna designs without analyzing workflow completeness, while Ramahatla et al. [?] detailed switching mechanisms for 5G/CubeSat applications without examining deployment pathway completeness.

Validation comprehensiveness functions as architectural decision point rather than passive verification. The three studies achieving five-dimensional integration [?], [?], [?]
—only 13% of feeding element corpus—employed multi-domain validation from initial design phases, enabling concurrent

electromagnetic-mechanical-thermal co-optimization. Studies with ≥ 3 validation domains achieved median integration scores of 4.2/5.0 versus 1.8/5.0 for single-domain approaches, indicating validation strategy predicts 48% of integration maturity variance. This converges with Abulgasem et al.'s [?] findings that 40% employed anechoic chamber and 30% VNA-based validation, confirming field-wide preference for accessible single-domain methodologies.

Root Cause Analysis: Structural Rather Than Technical. The persistence of systematic gaps across two decades despite advancing computational capabilities indicates structural causation. Three mechanisms emerge: (1) *Academic incentive misalignment*—electromagnetic novelty enables publication without thermal-mechanical validation, whereas comprehensive integration adds cost without proportional publication value. Full integration concentrates exclusively within satellite heritage programs (HORYU-IV, BIRDS-1, $n = 3/3$, 100%) where operational constraints override publication metrics. Hamid and Hock [?] documented exponential publication growth focusing on “microstrip patch antenna” and “3D fabrication,” yet minimal research connecting cost-effective techniques like xurography with antenna development. (2) *Disciplinary expertise fragmentation*—electromagnetic simulation expertise resides within antenna groups while thermal-mechanical qualification concentrates in spacecraft systems departments. Hamid and Hock's [?] bibliometric clustering confirmed “microstrip antennas” (1,395 occurrences) formed distinct clusters from “metamaterials” and “3D printers,” with xurography displaying limited connectivity to antenna research themes. (3) *Resource accessibility constraints*—the 70/30 distribution favoring accessible methodologies reflects infrastructure limitations. Specialized validation equipment (Satimo Starlab: 6%; TVAC: 14%) concentrates in institutions with active flight programs.

Mission-Tier-Specific Integration. Critical nuance: the 87% non-full-integration rate comprises both appropriate streamlining and problematic gaps. Technology demonstration missions (<1-year LEO) rationally avoid costly thermal qualification, whereas operational constellations (5+ years) require comprehensive validation. Hamid and Hock [?] identified three fabrication paradigms: PCB Prototyping (46.2%, median 6.5 dBi) suits near-term missions; Advanced Technologies (30.8%, median 11.0 dBi) target demanding applications but face qualification hurdles; Cost-Effective strategies (23.1%, median 7.5 dBi) support constellation deployment.

B. METHODOLOGICAL STRENGTHS AND CONTRIBUTIONS

This review's inductive empirical workflow configuration approach—analyzing how researchers organize development processes rather than imposing topology-based taxonomies—enabled data-driven identification of systematic discontinuities invisible to conventional reviews. Where taxonomy-focused approaches [?], [?], [?] effectively categorize antenna architectures, revealing frequency-specific performance characteristics, this workflow-based methodology exposes process-level gaps affecting deployment readiness

across all antenna categories.

Hamid and Hock [?] employed complementary bibliometric methodology using VOSviewer on 2,545 Scopus articles (2000–2021), revealing that while core electromagnetic concepts cluster strongly, rapid prototyping techniques demonstrate weak connectivity—corroborating that fabrication methodologies remain decoupled from antenna design discourse. The complementary nature becomes evident: topology reviews illuminate what designs are pursued, workflow analysis reveals why 92% fail flight qualification, and bibliometric analysis exposes research network structures perpetuating these gaps.

The 95% corpus coverage, rigorous PRISMA 2020 adherence, and convergent engineering-evidence synthesis provide robust empirical foundation. The documented attrition cascade (68%→87%→92%) emerges from systematic cross-study patterns with findings grounded in $n = 50$ (simulation/validation) and $n = 24$ (feeding integration) cohorts. The feeding-to-system gap (87%) and thermal validation deficiency (86%) represent empirically derived discoveries—quantitative findings not available in previous descriptive reviews.

C. IMPLICATIONS FOR PRACTICE AND FUTURE RESEARCH

Addressing workflow discontinuities requires multi-level intervention: (1) *For researchers*—multi-domain validation protocols should constitute baseline methodology for studies claiming deployment relevance. Mohammed et al.'s [?] comprehensive survey and Ramahatla et al.'s [?] examination of reconfiguration mechanisms provide technical foundations, yet our findings indicate these advances require systematic integration with multi-domain validation and thermal-mechanical characterization to achieve deployment readiness. Hamid and Hock [?] demonstrated hybrid fabrication approaches combining xurography with 3D-printed substrates offer accessible alternatives reducing hazardous waste. (2) *For satellite programs*—findings validate concurrent engineering wherein electromagnetic-mechanical-thermal constraints co-evolve from initial stages. Comprehensive validation concentrates exclusively in satellite heritage programs (HORYU-IV, BIRDS-1), indicating organizational structure influences workflow completeness. (3) *For funding agencies*—incentive realignment toward deployment readiness metrics may accelerate prototype-to-operational transition. Specialized validation equipment remains underutilized due to infrastructure barriers, suggesting targeted funding for shared facilities could democratize access.

These recommendations converge with Abulgasem et al.'s [?] future research priorities (miniaturization, multi-band operation, increased data rates) while extending discourse to systematic process maturation. Hamid and Hock's [?] bibliometric analysis revealed emerging clusters around meta-materials and flexible substrates, yet identified critical gaps where cost-effective techniques remain disconnected from mainstream fabrication networks. Most published designs

represent TRL 3–4 (proof-of-concept) rather than TRL 7–8 (flight-qualified).

As CubeSat missions expand toward operational constellation services, systematic workflows ensuring mission-ready antenna reliability become essential. The convergence between electromagnetic performance advancement (documented by Mohammed et al. [?]; Ramahatla et al. [?]) and process systematization (quantified in this review) represents the critical frontier for transitioning CubeSat antennas from experimental demonstrations to operational infrastructure. Future research should prioritize hybrid prototyping methods integrating accessible techniques, additive manufacturing, and comprehensive multi-domain validation to bridge documented workflow discontinuities while reducing environmental impact and democratizing fabrication capabilities.

V. CONCLUSIONS

This systematic review of 79 CubeSat antenna studies reveals systematic workflow discontinuities wherein extraordinary electromagnetic sophistication systematically decouples from deployment-ready integration, manifesting as a progressive attrition cascade (68%→70%→87%→92%) from simulation through fabrication. The empirical workflow configuration methodology—analyzing how researchers organize development processes rather than imposing predetermined categorization schemes—enables quantification of specific integration deficits invisible to topology-focused reviews, establishing evidence-based foundations for transitioning CubeSat antennas from experimental demonstrations toward operational infrastructure.

Electromagnetic Simulation-Optimization Integration.

The relationship between electromagnetic simulation platforms and design optimization recommendations demonstrates robust technical maturity yet profound documentation deficiencies. Three commercial solvers—Altair FEKO (34%), CST Microwave Studio (30%), and Ansys HFSS (26%)—collectively dominate 90% of implementations, achieving measured-to-simulated gain agreements within ± 1.5 dB across bandwidth enhancement, miniaturization, circular polarization, and gain maximization objectives. However, 68% omit critical algorithmic parameters (population sizes, crossover probabilities, convergence criteria), while cross-platform validation appears in fewer than 8% of studies, introducing vulnerabilities where solver-specific artifacts propagate without independent verification. This simulation ecosystem establishes a foundational constraint: incomplete computational documentation prevents experimental validation protocols from distinguishing solver artifacts from physical phenomena, ultimately propagating into manufacturing qualification gaps wherein fabrication teams lack computational benchmarks for acceptable tolerance ranges.

Experimental Validation-Protocol Alignment. Experimental validation configurations demonstrate natural complementarity between measurement instruments and validation practices, yet expose systematic gaps between laboratory characterization and operational verification. Four dominant

configurations emerged: anechoic chamber measurements (40%) emphasizing far-field patterns, VNA-based testing (30%) focusing on impedance characterization, specialized systems (16%) incorporating TVAC and near-field capabilities, and combined methodologies (14%) achieving comprehensive multi-domain validation. However, 70% of studies employ single-domain approaches inadequate for mission-ready deployment, with environmental testing underrepresented despite documented thermal-induced frequency shifts (± 8 MHz) and gain degradation (0.5–0.8 dB). Critically, validation strategy functions as architectural decision point rather than passive verification: studies employing multi-domain approaches achieved median integration scores of 4.2/5.0 versus 1.8/5.0 for single-domain methods, indicating validation comprehensiveness predicts 48% of integration maturity variance and determines whether designs progress toward deployment readiness or remain laboratory demonstrations.

Feeding Element Integration and System-Level Discontinuities. The integration of feeding elements exposes the review's most pronounced discontinuity—an 87% non-full-integration rate revealing systematic decoupling between component-level electromagnetic optimization and system-level deployment requirements. While electromagnetic simulation achieved universal coverage (100%, $n = 24$), critical downstream stages exhibited severe deficits: mechanical integration appeared in only 21% of studies, thermal analysis in 14%, and CubeSat body interaction modeling in 29%. This 71-percentage-point attrition from simulation to system integration—termed the “feeding-to-system gap”—concentrates complete five-dimensional integration exclusively within satellite heritage programs (HORYU-IV, BIRDS-1, $n = 3/3$, 100%) demonstrating concurrent engineering approaches wherein thermal-mechanical constraints inform electromagnetic optimization from initial design stages. The persistence of simplified feeding architectures (71% single-feed and coaxial/probe configurations) despite dramatic performance improvements (7.29 to 32.7 dBi gain) indicates that deployment reliability systematically outweighs electromagnetic performance optimization, with gain improvements deriving primarily from material advancements and geometric optimization rather than feeding network innovation.

Fabrication Workflow Configurations and Scalability Trade-offs. Three fabrication paradigms emerged with distinct performance-complexity-scalability profiles: PCB Prototyping (46.2%, median 6.5 dBi) provides accessible rapid prototyping yet exhibits tolerance sensitivity; Advanced Technologies (30.8%, median 11.0 dBi) achieve $1.7\times$ performance improvement through stacked substrates, 3D printing, and AMC integration but face qualification complexity; Cost-Effective strategies (23.1%, median 7.5 dBi) balance competitive performance with manufacturing scalability through simplified multilayer structures. However, systematic documentation gaps persist across paradigms: 46% omit bandwidth characterization, 92% lack environmental validation (only Curreli et al. [?] comprehensive), tolerance effects re-

ceive sparse attention, and zero studies report quantitative scalability analysis despite cost-effectiveness claims. These fabrication limitations directly reflect upstream integration deficits—tolerance assessment requires baseline thermal integration data present in only 14% of feeding designs—indicating fabrication workflow limitations stem from upstream discontinuities rather than manufacturing engineering deficiencies.

Limitations. This review confronts several methodological constraints. The exclusive focus on microstrip/patch/printed antenna technologies excludes alternative architectures (helix, monopole, deployable reflectors) potentially exhibiting different workflow maturation patterns. The 12 inaccessible studies (13% retrieval failure) may introduce selection bias if unavailable works disproportionately represent complete integration workflows. Workflow maturity assessment relies on reported methodologies; actual practices may exceed documentation levels, suggesting quantified gaps represent conservative lower bounds. The empirical workflow configuration approach, while revealing process-level discontinuities, provides limited prescriptive guidance for specific antenna topologies or frequency bands. Finally, temporal coverage (2000–2025) captures two decades of evolution yet may not fully represent emerging paradigms including AI-driven optimization, digital twins, and autonomous orbital assembly that could fundamentally restructure development workflows.

Research Gaps and Future Directions. Critical research imperatives emerge across four dimensions. First, multiphysics validation protocols integrating electromagnetic, thermal, mechanical, and deployment dynamics characterization from initial design stages require systematic development, particularly standardized TVAC testing procedures, radiation exposure qualification frameworks, and deployment mechanism reliability assessment methodologies addressing the documented 86% thermal validation deficiency. Second, feeding-to-beamforming integration pathways demand investigation, specifically examining how feeding network complexity influences system-level integration maturity, developing design methodologies wherein mechanical-thermal constraints inform electromagnetic optimization *ab initio*, and establishing quantitative relationships between feeding architecture and deployment readiness metrics. Third, fabrication process standardization and tolerance budgeting necessitate systematic tolerance-performance correlation studies, cross-paradigm manufacturing qualification protocols, and scalability assessment frameworks enabling cost-performance trade-off analysis for constellation deployment. Fourth, mission-tier-specific workflow templates require development, distinguishing appropriate integration levels for technology demonstration missions (<1 -year LEO) versus operational constellations (5+ years), preventing both underqualification of critical systems and over-engineering of demonstration platforms. As CubeSat missions transition from experimental demonstrations toward operational constellation services, bridging these documented workflow discontinuities through concurrent engineering paradigm

shifts—from isolated electromagnetic optimization toward integrated system modeling—becomes essential for ensuring antenna subsystem reliability and sustainable space system development.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this study. This study did not receive any financial support.

DATA AVAILABILITY

The complete research dataset supporting this study’s findings is available upon reasonable request for academic purposes. Materials include: trained neural network models, satellite imagery from 92 rural localities, geospatial-demographic databases, demand estimation calculations, LinkPlanner network design files, and the 3-million pixel annotated training dataset. All data are organized following FAIR principles with comprehensive metadata. Requests should be directed to sseguraal@unprg.edu.pe with intended use description. Data sharing agreements may apply to ensure compliance with institutional policies.

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S. FRANCISCO SEGURA He received a bachelor’s degree in Electronic Engineering from the National University of San Marcos in Lima, Peru, in 1996. He completed his master’s studies in Telecommunications and Networking at the Technological University of Peru, Lima, in 2021, and obtained another master’s degree in Applied Mathematics from the National University Pedro Ruiz Gallo, Lambayeque, Peru.

From 1996 to 2003, he worked as a Technical Manager for telecommunications operators, carrying out activities related to the operation and maintenance of broadband optical networks. In 2004, he joined as an Assistant Professor in the Department of Electronic Engineering at the National University Pedro Ruiz Gallo. Currently, he is an Associate Professor in the areas of telecommunications, broadband networks, image processing, and machine learning.

He is the author of articles on image processing, broadband networks, and machine learning.



LUCIA ISABEL CHAMAN CABRERA In 2004, after graduating with a degree in Electronic Engineering from the Private University Antenor Orrego, Trujillo, an individual embarked on a 12-year career in telecommunications, focusing on network design, operation, and maintenance. This practical experience laid the groundwork for a series of academic advancements, including a Master’s in Business Administration from the National University Pedro Ruiz Gallo in 2016, a Doctorate in Education from Cesar Vallejo University in 2021, and a Master’s in Educational Psychology from the same university in 2023. Transitioning into academia in 2013 as an auxiliary, and now associate professor in Electronic Engineering at the National University Pedro Ruiz Gallo, this person has applied a deep technical and pedagogical knowledge to educate future engineers. Research interests have evolved from signal processing and deep learning to educational psychology, aiming to enhance teaching efficacy. This biography highlights a professional dedicated to merging technical prowess with educational insight, significantly impacting engineering education and research. Through continuous learning and innovation, this individual contributes profoundly to both fields, exemplifying a commitment to societal advancement through education and technology.



DIANA M. CASTRO CÁRDENAS Ph.D., is a Principal Lecturer in the Department of Mathematics at the Faculty of Physical Sciences and Mathematics, National University Pedro Ruiz Gallo. She earned her Bachelor’s degree in Mathematics from the National University San Luis Gonzaga de Ica in 1993, followed by a Master’s degree in Applied Mathematics from the National University Pedro Ruiz Gallo in 2016, and a Doctorate in Education from the University Cesar Vallejo in 2020. Her

research interests encompass flipped classrooms, neural networks, finite elements, numerical methods, and optimization.

APPENDIX A DETAILED HYPERPARAMETERS CONFIGURATION

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